
Compact Sensor Array Design and Verification

Analytical Development, Numerical Modeling, and Engineering
Evidence

Applied Electromagnetics and Embedded Sensing

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Independent Engineering Portfolio Project

Repository: <https://github.com/username/project-repository>

July 25, 2026

Portfolio Context and Evidence

Project origin. This public edition was prepared from an independently completed engineering project. Replace this sentence with a concise, factual description of the project origin.

Public portfolio edition. Administrative, grading, student-identification, and institutional material has been removed. The public edition focuses on technical decisions, evidence, limitations, and reproducibility.

Analytical Evidence

Closed-form calculations, governing equations, design rules, and independent scripts.

Simulated Evidence

Numerical model construction, solver settings, parameter sweeps, and exported results.

Measured Evidence

No fabricated prototype or measurement campaign is claimed in this example template.

Limitation or Qualification

Replace this paragraph with solver, mesh, boundary, material, convergence, and software limitations that materially affect interpretation of the reported results.

Affiliation disclaimer. This document is an independent portfolio artifact and does not imply endorsement by an employer, university, instructor, customer, software vendor, or standards organization.

Prepared as a public engineering portfolio artifact.

Abstract

This template demonstrates how to convert a completed engineering project into a public, evidence-based portfolio report. The example narrative separates analytical work, numerical simulation, and measured evidence; records assumptions and limitations; and connects each public claim to a reproducible source. Replace this sample abstract with a concise statement of the engineering problem, method, principal verified results, and practical significance. Avoid unqualified performance claims and clearly distinguish simulated behavior from measured hardware performance.

Project Objective

The objective of this portfolio report is to document a complete engineering workflow from problem definition through analytical design, model construction, numerical or experimental verification, interpretation, and public claim control. The final report should allow a technically competent reviewer to understand what was designed, why specific decisions were made, what evidence supports the conclusions, and which limitations remain unresolved.

Engineering Decision

A portfolio report is not merely a shortened academic submission. It is a public engineering artifact organized around judgment, evidence, reproducibility, and honest qualification of uncertainty.

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Chapter 1

Project Overview

1.1 Engineering problem

State the practical engineering problem in terms a hiring manager, customer, or technical reviewer can evaluate. Identify the operating environment, performance objective, constraints, and consequences of failure. Replace generic phrases such as “optimize the system” with measurable engineering targets.

1.2 System context

Figure 1.1 shows the recommended evidence flow for a public engineering project. The report begins with requirements, proceeds through analytical and numerical work, and ends with controlled public claims.

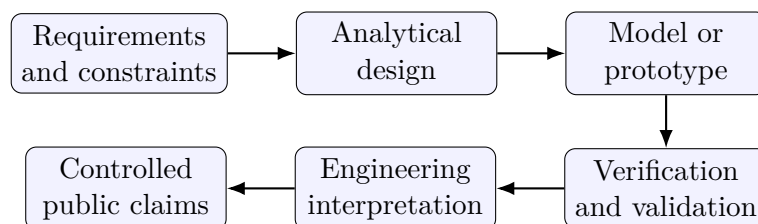


Figure 1.1: Recommended evidence flow for an engineering portfolio project.

1.3 Portfolio value

Explain what this project demonstrates beyond software operation. Examples include requirements translation, model credibility, trade-space reasoning, numerical reliability,

root-cause analysis, design iteration, verification discipline, and technical communication.

Chapter 2

Engineering Scope and Evidence

2.1 Scope boundaries

Define what is included and excluded. A strong scope statement prevents the public report from implying fabrication, certification, field validation, or production readiness that was not actually achieved.

Table 2.1: Example scope boundary table.

Area	Included	Excluded or deferred
Analytical work	Governing equations, first-order sizing, sensitivity checks	Formal uncertainty propagation
Numerical work	Baseline model, parameter sweep, selected convergence checks	Full certification-grade validation
Hardware	Interface and test concept	Fabricated prototype and chamber measurements
Public release	Verified plots, scripts, assumptions, limitations	Proprietary files and institution-specific administration

2.2 Evidence taxonomy

Use evidence labels consistently throughout the report:

- **ANALYTICAL** results derived from equations, calculations, or independent scripts;
- **SIMULATED** results exported from a numerical model or digital twin;

- **MEASURED** results obtained from physical instrumentation or test equipment;
- **QUALIFIED** claims that remain dependent on stated assumptions or incomplete validation.

2.3 Deliverables

List the public artifacts that allow a reviewer to reproduce or audit the work: source report, code, console output, model screenshots, run manifest, parameter tables, verification matrix, and selected native project files when licensing and size permit.

Chapter 3

Theory and Analytical Design

3.1 Governing model

Introduce only the theory needed to justify engineering choices. For antenna and array projects, canonical references such as Stutzman and Thiele provide a disciplined starting point for notation and first-order design relationships [1]. Define variables, assumptions, coordinate systems, and validity limits. For an illustrative normalized array or sensing problem, a generic response may be written as

$$H(\theta) = \sum_{n=0}^{N-1} w_n e^{jn\psi(\theta)}, \quad (3.1)$$

where w_n represents a complex weighting coefficient and $\psi(\theta)$ represents the progressive phase associated with geometry and excitation.

3.2 First-order design calculation

Show the calculation chain that converts requirements into initial dimensions or parameters. The purpose is not to reproduce a textbook chapter; it is to establish a traceable starting point for modeling and optimization.

Analytical Evidence

The example script in Appendix A calculates a normalized wavelength, nominal spacing, and progressive phase. Replace it with the project's real verification script and retain its console output under `verification_records/`.

3.3 Assumptions and validity

Record assumptions before presenting results. Typical examples include linearity, steady-state operation, homogeneous materials, ideal source behavior, negligible connector effects, limited temperature range, or simplified boundary conditions.

Chapter 4

Model and Simulation Method

4.1 Model construction

Describe geometry, material properties, excitations, ports, boundary conditions, monitors, mesh controls, solver type, stopping criteria, and post-processing. A second engineer should be able to reconstruct the model from the information presented here and in the simulation record.

4.2 Parameter control

Maintain a single source of truth for design parameters. Table 4.1 illustrates a compact public parameter record.

Table 4.1: Example public parameter record.

Parameter	Symbol	Example value	Evidence source
Reference frequency	f_0	2.50 GHz	Requirement
Normalized wavelength	λ_0	119.9 mm	Analytical script
Nominal spacing	d	$0.5\lambda_0$	Analytical design
Relative phase	α	90°	Steering case

4.3 Run manifest

Record each meaningful numerical run with a stable identifier, model revision, solver, mesh or discretization setting, changed parameter, completion status, and exported

artifacts. The starter file `simulation_records/run_manifest.csv` provides a machine-readable structure.

Simulated Evidence

A screenshot alone is not a complete simulation record. Pair every published plot with a run identifier and enough configuration information to reconstruct its origin.

Chapter 5

Results and Verification

5.1 Result presentation

Present the minimum set of figures and tables needed to support the engineering conclusion. Each caption should identify the evidence class, operating condition, and relevant run or test identifier.

Table 5.1: Example comparison of analytical and numerical evidence.

Metric		Analytical		Simulated		Interpretation
Characteristic frequency	fre-	Requirement-derived		Solver-extracted		Agreement must be quantified
	Primary response	Idealized model		Full numerical model		Differences require physical explanation
Efficiency or loss		Simplified estimate	esti-	Material and solver dependent		State model fidelity
Steering or control		Closed-form prediction	pre-	Post-processed result		Report scan loss and asymmetry

5.2 Verification checks

Verification asks whether the equations, scripts, model, and post-processing were implemented correctly. Useful checks include unit tests, hand calculations, independent scripts, reciprocity, symmetry, conservation, limiting cases, and comparison with canonical results.

5.3 Validation status

Validation asks whether the model represents physical reality adequately for the intended use. Explicitly distinguish model verification from experimental validation.

Measured Evidence

This starter report claims no measured evidence. Replace this box only after a real prototype, calibrated setup, test procedure, uncertainty statement, and traceable data record exist.

Chapter 6

Engineering Discussion

6.1 Interpretation rather than description

Do not merely restate that a curve moved or a field pattern changed. Explain the physical or algorithmic mechanism, identify the controlling parameter, and connect the result to requirements and design risk.

6.2 Trade-offs

Engineering portfolios become credible when trade-offs are explicit. Discuss competing objectives such as bandwidth versus size, accuracy versus runtime, isolation versus spacing, latency versus resource use, robustness versus complexity, and maintainability versus optimization.

6.3 Numerical reliability and limitations

Document mesh or discretization adequacy, boundary sensitivity, solver tolerance, material fidelity, numerical noise, anomalous behavior, and convergence evidence. Verification and validation should be treated as distinct technical activities rather than interchangeable labels [2], [3].

Limitation or Qualification

A visually smooth plot is not proof of numerical reliability. Non-monotonic or unexpected behavior should trigger mesh, boundary, solver, or implementation checks. When those checks are not feasible, state the limitation directly and narrow

the claim.

6.4 Industrial relevance

Translate the technical result into a design, integration, manufacturing, maintenance, safety, cost, or deployment consequence. This section should demonstrate engineering judgment rather than marketing language.

Chapter 7

Engineering Verification Matrix

The verification matrix is the claim-control center of the public report. Every major claim should identify its evidence class, source, and limitation. Replace the sample rows with project-specific entries before publication.

Table 7.1: Engineering verification matrix.

ID	Public claim	Evidence class	Evidence location	Limitation or qualification
V1	Initial dimensions or parameters were calculated independently.	Analytical	Equation (3.1); Appendix A	Depends on stated first-order assumptions.
V2	The numerical model reproduces the documented baseline configuration.	Simulated	Model-method chapter; run manifest	Requires project-specific solver and mesh records.
V3	The reported trend is physically interpreted and compared with the analytical expectation.	Analytical + Simulated	Results and discussion chapters	Does not constitute experimental validation.
V4	No measured hardware performance is claimed in the starter document.	Measured status	Portfolio Context and Evidence	Replace only after traceable physical testing.

Chapter 8

Conclusion

Summarize the verified engineering outcome, not the sequence of report sections. State which requirements were met, which evidence supports that conclusion, what trade-offs were accepted, and what remains unverified. Keep analytical, simulated, and measured claims separate.

A strong final paragraph identifies the next technically defensible step, such as mesh or boundary convergence, prototype fabrication, chamber measurement, hardware-in-the-loop testing, environmental qualification, sensitivity analysis, or production design review.

Appendix A

Analytical Verification Script

This appendix demonstrates how to preserve an independent calculation inside the public source package. Replace the starter script with the actual project calculation and commit the corresponding console output.

```
1  """Starter analytical verification for the portfolio report template."""
2
3  from __future__ import annotations
4
5  from dataclasses import dataclass
6  from math import pi
7
8  CO_M_PER_S = 299_792_458.0
9
10
11 @dataclass(frozen=True)
12 class DesignCase:
13     frequency_hz: float
14     spacing_fraction: float
15     steering_angle_deg: float
16
17     def validate(self) -> None:
18         if self.frequency_hz <= 0:
19             raise ValueError("frequency_hz must be positive")
20         if not 0 < self.spacing_fraction <= 1:
21             raise ValueError("spacing_fraction must be in (0, 1]")
22         if not -90 <= self.steering_angle_deg <= 90:
23             raise ValueError("steering_angle_deg must be within [-90, 90]")
24
25
26 def calculate(case: DesignCase) -> dict[str, float]:
27     case.validate()
28     wavelength_m = CO_M_PER_S / case.frequency_hz
29     spacing_m = case.spacing_fraction * wavelength_m
30     steering_angle_rad = case.steering_angle_deg * pi / 180.0
31     phase_rad = -2.0 * pi * spacing_m / wavelength_m * __import__("math").sin(
```

```
32     steering_angle_rad
33 )
34 return {
35     "wavelength_m": wavelength_m,
36     "spacing_m": spacing_m,
37     "phase_rad": phase_rad,
38     "phase_deg": phase_rad * 180.0 / pi,
39 }
40
41
42 def main() -> None:
43     case = DesignCase(
44         frequency_hz=2.50e9,
45         spacing_fraction=0.50,
46         steering_angle_deg=30.0,
47     )
48     result = calculate(case)
49     print(f"Frequency: {case.frequency_hz / 1e9:.3f} GHz")
50     print(f"Wavelength: {result['wavelength_m'] * 1e3:.3f} mm")
51     print(f"Spacing: {result['spacing_m'] * 1e3:.3f} mm")
52     print(f"Progressive phase: {result['phase_deg']:.3f} deg")
53
54
55 if __name__ == "__main__":
56     main()
```

Listing A.1: Starter analytical verification script.

```
Frequency: 2.500 GHz
Wavelength: 119.917 mm
Spacing: 59.958 mm
Progressive phase: -90.000 deg
```

Listing A.2: Recorded console output from the starter script.

Appendix B

Simulation and Model Record

For every published numerical result, retain a stable record containing the following:

- run identifier and model revision;
- software and solver version;
- geometry and material parameter set;
- excitation, port, and boundary configuration;
- mesh or discretization settings;
- convergence or stopping criteria;
- exported plots, tables, and raw data;
- known warnings, simplifications, and failed checks.

Table B.1: Example simulation-run record.

Field	Example entry
Run ID	SIM-001
Purpose	Baseline model before parameter sweep
Model revision	rev-A
Solver	Replace with solver and version
Discretization	Replace with cell count, mesh rule, or element order
Convergence	Replace with tolerance and achieved status
Artifacts	Plot names, raw exports, and screenshots
Limitations	Material, boundary, license, runtime, or memory constraints

References

- [1] W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed. John Wiley & Sons, 2012.
- [2] W. L. Oberkampf and C. J. Roy, *Verification and Validation in Scientific Computing*. Cambridge University Press, 2010.
- [3] *Standard for verification and validation in computational fluid dynamics and heat transfer*, V&V 20-2009, ASME, 2009.